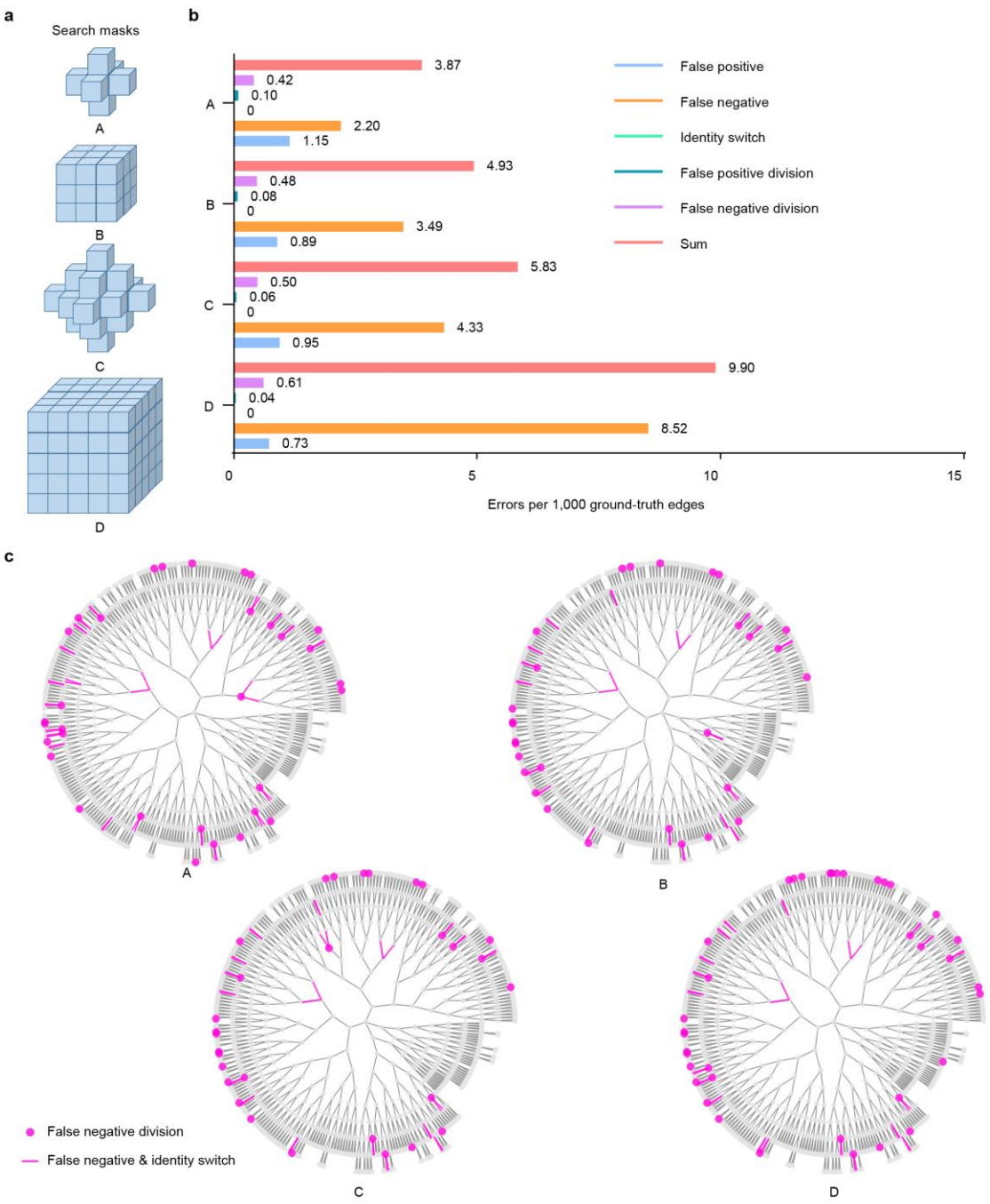


Supplementary Figure 10 | Influence of the search-mark structure on tracking performance. The impact of search mask structure on the model's performance was evaluated, revealing that performance is predominantly influenced by cell density and axial resolution. Among the 4 different search mask structures assessed, ensuring the absence of multiple center points within a mask's range was a key consideration. However, with very low axial resolution, larger search masks occasionally failed to detect certain center points. To address this, we adopted the most streamlined structure A. Error connections are highlighted in pink, while correct connections are depicted in grey. All evaluation data in this figure were drawn from mskcc-confocal dataset #2. All models were trained using mskcc-confocal dataset #3.